CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2017 Examination | **Paper Code:** 18023 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

### Q1. Describe the various components of Data Communication. [3 Marks]

**Answer:**

Data communication is the exchange of data between two devices via a transmission medium. It comprises five essential components:

| Component | Description & Role |
|---|---|
| **1. Message** | The information or payload to be communicated (Text, Numbers, Pictures, Audio, Video). |
| **2. Sender (Transmitter)** | The source device originating and transmitting the data message (PC, Workstation, Smartphone). |
| **3. Receiver** | The destination device that receives the transmitted data (Server, Printer, Mobile terminal). |
| **4. Transmission Medium** | The physical communication conduit carrying the signal from sender to receiver: Guided (Twisted pair, Coaxial cable, Fiber optic) or Unguided (Radio waves, Microwaves, Infrared). |
| **5. Protocol** | A formal set of governing rules that coordinate data transmission syntax, semantics, and timing (e.g., TCP/IP, HTTP). |

### Q2. Define Bandwidth and Network Congestion. [3 Marks]

**Answer:**

**1. Bandwidth:**
The maximum data transfer capacity of a communication channel over a given interval. In digital systems, it is measured in **bits per second (bps, Mbps, Gbps)**; in analog systems, it represents the frequency range (in Hertz, Hz) between upper and lower cutoff frequencies ( $Bandwidth = f_{high} - f_{low}$ ).

**2. Network Congestion:**
A state where network nodes (routers/switches) carry more data packets than they can accommodate. When offered traffic load exceeds channel capacity, router queues overflow, causing **packet queuing delay, packet loss, and throughput collapse**.

### Q3. Distinguish between MAC Address and IP Address. [3 Marks]

**Answer:**

| Parameter | MAC Address (Physical Address) | IP Address (Logical Address) |
|---|---|---|
| **OSI Layer** | Data Link Layer (Layer 2). | Network Layer (Layer 3). |
| **Address Length & Format** | **48 bits (6 bytes)**; written in 12 hexadecimal digits (e.g., `00:1A:2B:3C:4D:5E`). | **32 bits** in IPv4 (e.g., `192.168.1.1`); **128 bits** in IPv6 (hexadecimal). |
| **Permanence & Assignment** | Permanently hard-coded into Network Interface Card (NIC) by manufacturer (OUI). | Dynamically (DHCP) or statically assigned based on network location. |
| **Scope of Utility** | Local node-to-node frame delivery within the same LAN segment. | Global end-to-end packet routing across multiple networks/Internet. |

### Q4. Differentiate between a Hub and a Switch. [3 Marks]

**Answer:**

| Feature | Network Hub | Network Switch |
|---|---|---|
| **Operating OSI Layer** | Physical Layer (Layer 1). | Data Link Layer (Layer 2) / Multilayer (Layer 3). |
| **Data Transmission Mode** | **Broadcasts** all incoming frames to every connected port indiscriminately. | **Unicasts** frames selectively to the specific destination port using MAC table. |
| **Collision Domains** | All ports share a single collision domain (high collision risk). | Each individual port forms its own dedicated collision domain. |
| **Bandwidth Sharing** | Total bandwidth is shared among all connected active nodes. | Dedicated, full-rate bandwidth provided to each port (full-duplex). |

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### Q5. What is Subnetting? Give an example with CIDR calculation. [3 Marks]

**Answer:**

**Subnetting** is the practice of logically partitioning a single large IP network into smaller, distinct subnetworks (subnets) by borrowing host bits for the network ID. This reduces broadcast domains, improves routing efficiency, and conserves IP addresses.

**Example: Subnetting 192.168.1.0/24 into 4 Subnets**

- To create 4 subnets: Borrow 2 bits from host portion ( $2^2 = 4$ subnets).
- New prefix length: `/24 + 2 = /26` ⇒ Subnet Mask: `255.255.255.192`.
- Block Size: `256 - 192 = 64` addresses per subnet (62 usable hosts).

| Subnet # | Network ID | Usable Host IP Range | Directed Broadcast IP |
|---|---|---|---|
| Subnet 1 | `192.168.1.0` | `192.168.1.1 - 192.168.1.62` | `192.168.1.63` |
| Subnet 2 | `192.168.1.64` | `192.168.1.65 - 192.168.1.126` | `192.168.1.127` |
| Subnet 3 | `192.168.1.128` | `192.168.1.129 - 192.168.1.190` | `192.168.1.191` |
| Subnet 4 | `192.168.1.192` | `192.168.1.193 - 192.168.1.254` | `192.168.1.255` |

## SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

### Q6. Describe the role of the Transport Layer in the OSI Model. [7.5 Marks]

**Answer:**

The **Transport Layer (Layer 4)** serves as the boundary between hardware-oriented subnet layers and software application layers, providing true **end-to-end, process-to-process reliable data delivery**.

| Core Function | Detailed Mechanism & Protocol Realization |
|---|---|
| **1. Service-Point (Port) Addressing** | Differentiates concurrent application processes executing on the same host using 16-bit Port numbers (e.g., HTTP: 80, HTTPS: 443, DNS: 53, SSH: 22). |
| **2. Segmentation & Reassembly** | Splits massive application messages into manageable chunks called **Segments**, appends sequence numbers, and reassembles them at receiver in exact order. |
| **3. Connection Management** | Manages connection-oriented transmission via 3-way handshake (TCP) or fast connectionless datagram delivery (UDP). |
| **4. Flow Control** | Uses **Sliding Window Protocol** to prevent a fast transmitter from overwhelming a slow receiver's buffer space. |
| **5. Error Control** | Enforces end-to-end integrity using 16-bit **Checksums** and Automatic Repeat reQuest (ARQ) retransmissions for dropped/corrupted segments. |

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**Q7. Explain Network Topologies (Star, Mesh, Bus, Ring) with diagrams and comparisons.** [7.5 Marks]

**Answer:**

**Network Topology** defines the geometric and logical arrangement of communication links and nodes in a network.

**1. Star Topology:**

All nodes connect via dedicated point-to-point links directly to a central multiport device (Hub or Switch).

*Layout:* `[Node A, Node B, Node C, Node D] → [Central Switch / Hub]`

**Pros:** Easy to install, failure of one node does not impact others. **Cons:** Central hub is a single point of failure.

**2. Mesh Topology (Full Mesh):**

Every node has a dedicated point-to-point link to every other node in the network.

*Link Formula:* For `n` nodes, requires $\frac{n(n-1)}{2}$ physical links and `(n - 1)` I/O ports per device.

**Pros:** Maximum redundancy, zero traffic contention, highly fault-tolerant. **Cons:** Extremely expensive cabling and port requirements.

**3. Bus Topology:**

All nodes share a single common linear backbone cable (coaxial cable with terminators at both ends).

*Layout:* `Terminator ── [Node 1] ── [Node 2] ── [Node 3] ── Terminator`

**Pros:** Minimal cabling needed. **Cons:** A single break in backbone takes down entire network; hard to troubleshoot.

**4. Ring Topology:**

Each device connects directly to its two immediate neighbors, forming a closed continuous loop.

*Layout:* `Node 1 → Node 2 → Node 3 → Node 4 → Node 1 (Token Passing)`

**Pros:** Orderly token transmission, no collisions. **Cons:** Break in ring disrupts network (unless dual counter-rotating ring like FDDI is used).

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**Q8. Explain Congestion Control Algorithms (Leaky Bucket and Token Bucket).** [7.5 Marks]

**Answer:**

Traffic shaping algorithms regulate the bursty flow of packets injected into the network to mitigate congestion:

| Parameter | Leaky Bucket Algorithm | Token Bucket Algorithm |
|---|---|---|
| **Core Operating Principle** | Bursty packet arrivals enter a finite buffer; packets leak out at a **strictly constant, rigid rate**. | Tokens generate into bucket at regular rate `r` up to capacity `C`. Packets transmit by consuming tokens. |
| **Bursty Traffic Handling** | **Smooths out all bursts completely**; discards packets if bucket capacity overflows. | **Allows controlled bursts** of packets up to the accumulated token capacity `C`. |
| **Output Transmission Rate** | Strictly uniform, deterministic constant rate regardless of burstiness. | Variable rate bounded by average rate `r` and peak burst capacity. |
| **Packet Discard Policy** | Discards incoming packets when buffer memory fills up. | Discards tokens when token bucket fills; packets queue or wait for fresh tokens. |

## SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

### Q9. Explain the TCP Three-Way Handshake with a sequence diagram. [15 Marks]

**Answer:**

The **TCP 3-Way Handshake** is the synchronization mechanism used by Transmission Control Protocol to establish a reliable, full-duplex virtual connection between client and server before data exchange begins.

**TCP 3-Way Handshake Connection Sequence:**

| Step | Direction | Control Flags & Sequence # | Action & Purpose |
|---|---|---|---|
| **Step 1 (SYN)** | Client → Server | `SYN = 1`<br>`Seq = X` (Client ISN) | Client requests connection and announces its initial sequence number X. Client enters **SYN-SENT** state. |
| **Step 2 (SYN-ACK)** | Server → Client | `SYN = 1, ACK = 1`<br>`Seq = Y` (Server ISN)<br>`Ack_Num = X + 1` | Server acknowledges client's request (`Ack = X + 1`) and provides its own sequence number Y. Server enters **SYN-RCVD**. |
| **Step 3 (ACK)** | Client → Server | `ACK = 1`<br>`Seq = X + 1`<br>`Ack_Num = Y + 1` | Client acknowledges server's sequence number (`Ack = Y + 1`). Connection enters **ESTABLISHED** state on both endpoints. |

**Significance:** Guarantees both parties agree on initial sequence numbers, confirms bidirectional communication capability, and prevents old duplicate connection requests from causing false sessions.

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### Q10. Draw and explain the 7 Layers of the ISO/OSI Reference Model. [15 Marks]

**Answer:**

The **Open Systems Interconnection (OSI)** reference model is a conceptual 7-layer architectural framework developed by ISO for standardized computer communication:

| Layer | Layer Name | PDU | Primary Functions & Responsibilities | Standard Protocols |
|---|---|---|---|---|
| **Layer 7** | **Application** | Data | Provides user interface and network service abstraction for software apps. | HTTP, HTTPS, FTP, SMTP, DNS, DHCP |
| **Layer 6** | **Presentation** | Data | Data syntax translation, compression, encryption and decryption (SSL/TLS). | JPEG, MPEG, ASCII, TLS/SSL |
| **Layer 5** | **Session** | Data | Dialog control, token management, session checkpointing and synchronization. | RPC, NetBIOS, PPTP |
| **Layer 4** | **Transport** | Segment | End-to-end process delivery, port addressing, flow control (sliding window), error recovery. | TCP, UDP |
| **Layer 3** | **Network** | Packet | Logical IP addressing, subnet routing, packet forwarding across networks. | IPv4, IPv6, ICMP, OSPF, BGP |
| **Layer 2** | **Data Link** | Frame | Node-to-node frame delivery, physical MAC addressing, CRC error detection, CSMA/CD. | Ethernet (802.3), Wi-Fi (802.11), PPP |
| **Layer 1** | **Physical** | Bit | Transmission of raw bitstream over physical media, signal encoding, voltages. | RS-232, RJ-45, V.35, Cables |

### Q11. Explain Shannon Capacity Theorem with a solved numerical problem. [15 Marks]

**Answer:**

**Shannon's Capacity Theorem** establishes the theoretical upper limit on the maximum data transmission rate (Channel Capacity C) across a noisy communication channel with white Gaussian thermal noise.

> Shannon Formula: $C = B \bullet \log_2 \&left;( 1 + \frac{S}{N} \&right;)$ (in bits per second, bps)
>
> Where:
> - `C` = Maximum theoretical channel capacity in bits per second (bps)
> - `B` = Channel bandwidth in Hertz (Hz)
> - `S / N` = Signal-to-Noise Ratio (SNR as a linear power ratio)

**Solved Numerical Problem:**

> **Problem Statement:** A noisy telephone channel has a bandwidth of *B = 3000 Hz* and a Signal-to-Noise Ratio of $SNR_{dB}$ *= 35 dB*. Calculate the theoretical maximum channel capacity.
>
> **Step 1: Convert SNR from decibels (dB) to linear ratio:**
>
> $$\text{SNR}_{\text{dB}} = 10 \bullet \log_{10}(S/N) = 35 \Rightarrow \log_{10}(S/N) = 3.5 \Rightarrow \frac{S}{N} = 10^{3.5} \approx 3162.28$$
>
> **Step 2: Apply Shannon's Formula:**
>
> $$C = 3000 \bullet \log_2(1 + 3162.28) = 3000 \bullet \log_2(3163.28)$$
>
> Using change of base: $\log_2(3163.28) = \frac{\log_{10}(3163.28)}{\log_{10}(2)} = \frac{3.50014}{0.30103} \approx 11.6272$
>
> $$C = 3000 \times 11.6272 \approx 34{,}881.6 \text{ bps} \approx 34.88 \text{ kbps}$$

**Theoretical Channel Capacity: C ≈ 34.88 kbps**

### Q12. Explain CSMA/CD in Ethernet and compare with CSMA/CA in Wi-Fi. [15 Marks]

**Answer:**

**Carrier Sense Multiple Access (CSMA)** protocols require nodes to "listen before talking" on a shared broadcast transmission medium.

| Comparison Feature | CSMA/CD (Collision Detection) | CSMA/CA (Collision Avoidance) |
|---|---|---|
| **Standard & Medium** | IEEE 802.3 (Wired Ethernet cables). | IEEE 802.11 (Wireless Wi-Fi radio channels). |
| **Collision Strategy** | Transmits while listening; if collision signal is detected, stops immediately. | Avoids collisions prior to transmission via reservation handshakes and delays. |
| **Collision Response** | Transmits a **Jam Signal** to notify all stations, then applies **Binary Exponential Backoff**. | Cannot detect collision while sending (half-duplex RF); infers collision if no ACK received. |
| **Control Mechanism** | Collision detection hardware listening for elevated cable voltage. | **RTS (Request to Send) / CTS (Clear to Send)** handshakes + Inter-Frame Spacing (DIFS/SIFS). |

### Q13. Compare Circuit Switching, Packet Switching, and Message Switching. [15 Marks]

**Answer:**

**Switching** is the methodology by which nodes communicate across multiple intermediate network nodes without direct point-to-point physical links.

| Feature | Circuit Switching | Packet Switching | Message Switching |
|---|---|---|---|
| **Dedicated Physical Path** | **Yes**; dedicated end-to-end copper/fiber circuit reserved for session duration. | **No**; dynamic independent packet routing over shared virtual links. | **No**; whole message hops from node to node across shared lines. |
| **Call Setup Phase** | Required prior to any data transmission. Adds initial setup latency. | Zero connection setup latency (in Connectionless Datagram mode). | No setup phase needed. |
| **Bandwidth Utilization** | Low/wasteful; reserved circuit capacity is idle during conversational silence. | **Very High**; dynamic statistical multiplexing shares channel bandwidth. | Moderate; links occupied only while whole message transmits. |
| **Store & Forward Mechanism** | None; continuous real-time bitstream. | Packets buffered at each router and forwarded. Minimal latency. | Entire message buffered on intermediate disks; very high delay. |
| **Primary Application** | Traditional Landline Telephony (PSTN), cellular voice calls. | **The Modern Internet (IP)**, VoIP, streaming, web browsing. | Telegraph, Telex, historic military communication systems. |